GRAPH CONSTRUCTION FROM DATA BY NON-NEGATIVE KERNEL REGRESSION

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ABSTRACT

Data driven graph constructions are often used in machine learning applications. However, learning an optimal graph from data is still a challenging task. K-nearest neighbor and ϵ-neighborhood methods are among the most common graph construction methods, due to their computational simplicity, but the choice of parameters such as K and ϵ associated with these methods is often ad hoc and lacks a clear interpretation. The main novelty of this paper is to formulate graph construction as the problem of finding a sparse signal approximation in kernel space, and identifying key similarities between methods in signal approximation and existing graph learning methods. We propose non-negative kernel regression (NNK), an improved approach for graph construction with interesting geometric and theoretical properties. We demonstrate experimentally the efficiency of NNK graphs, their robustness to choice of sparsity K and show that they can outperform state of the art graph methods in semi supervised learning tasks.

Index Terms— Graph learning, Graph construction, KNN graph, ϵ graph, LLE Graph, Neighborhood graph.

1. INTRODUCTION

Graph's ability to capture complex structure underlying high dimensional data has drawn interest from various fields such as machine learning [2] and signal processing [3]. In machine learning, graph methods have been applied to manifold embedding [4], spectral clustering [5] and semi supervised learning [6], and more recent applications include using graphs as structured regularizers or as tools to understand black box learning models [7, 8, 9]. In a graph construction or graph learning problem, we are given N data points with feature vectors $\{\boldsymbol{x}_1, \boldsymbol{x}_2 \ldots \boldsymbol{x}_N\} \in \mathbb{R}^d$ and the goal is to obtain an efficient graph representation, for example, one where the number of edges is of the same order as the number of nodes. Such sparse graph constructions lead to faster downstream graph processing in operations such as finding flow cuts, information propagation, and others [10]. The most popular graph constructions are weighted K-nearest neighbor (KNN) and ϵ-neighborhood graphs (ϵ-graph). Though these graphs exhibit manifold convergence properties [11, 12], finding an efficient graph representation given finite, non uniformly sampled data still remains an open and difficult task [13]. Algorithms previously proposed for solving graph learning can be broadly classified into two main families: *similarity-based* and *locality-inducing* [14].

Similarity based approaches compute a similarity metric between every pair of points and associate this value to the edge weight between the corresponding nodes. A commonly used metric derived from distance is the Gaussian kernel with variance (bandwidth) σ^2:

$$k_{\sigma^2}(\boldsymbol{x}_i, \boldsymbol{x}_j) = e^{-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2\sigma^2}}. \tag{1}$$

Methods in this class sparsify the *fully connected* graph to reduce complexity while seeking to maintain the properties of the original graph. KNN and ϵ-graphs are both examples of similarity based constructions. Some more recent similarity-based approaches use distance metrics learned from data. Examples include kernel learning [15] and adaptive edge weighting (AEW) [16]. These methods optimize the edge weights of KNN/ϵ-graphs while leaving connectivity unchanged. The main challenge with similarity-based graph constructions is that they can suffer from bias-variance issues related to sparsification parameters such as K and ϵ [17].

Locality inducing approaches compute edge weights around a given data point so as to better reflect local characteristics of data points. As an example, local linear embedding (LLE) [18] under constraints [19] solves:

$$\underset{\boldsymbol{\theta}:\ \boldsymbol{\theta} \geq 0}{\text{minimize}}\ ||\boldsymbol{x}_i - \boldsymbol{X}_S\boldsymbol{\theta}||_{l_2}^2, \tag{2}$$

where $\boldsymbol{X}_S$ is the matrix containing the nearest neighbors of $\boldsymbol{x}_i$ whose indices are denoted by set S. The solution $\boldsymbol{\theta}$ corresponds to the weights of the edges connecting the neighbors. Modifications to [19] include [20], which introduces b-matching to enforce regularity in graphs, and [21], which proposes a graph construction based on a *global* formulation of the LLE measure under positivity constraints. Recent methods proposed by Kalofolias et. al [22, 23] can be grouped under this class where the global objective of locality inducing methods is redefined from a graph signal perspective with explainable regularizers. Locality inducing methods assume the input data space is locally linear and are susceptible to noise [24].

Our approach is to view graph construction as a signal representation problem. Recall that a positive definite kernel $k(\boldsymbol{x}_i, \boldsymbol{x}_j)$, corresponds to a transformation of data points in $\mathbb{R}^d$ to points in a (possibly infinite dimensional) Hilbert space, such that similarities can be interpreted as dot products in this space, i.e., $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{\phi}_i^\top \boldsymbol{\phi}_j$, where ϕ is a mapping from data space to transformed space and $\boldsymbol{\phi}_i$ represents the transformed observation $\boldsymbol{x}_i$. This dot product space, or Reproducing Kernel Hilbert Space (RKHS), has well established properties and applications [25] but has not been sufficiently studied for the purpose of graph learning. We observe that KNN (ϵ-graph) can be construed from a signal approximation perspective as choosing the K largest inner products $\boldsymbol{\phi}_i^\top \boldsymbol{\phi}_j$ (inner products above threshold ϵ). Based on this observation, we propose non negative kernel regression (NNK), which can be viewed as an improved basis selection procedure for graph construction (NNK graph). The

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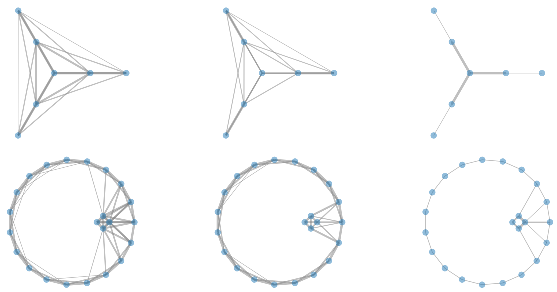


(a) KNN Graph (b) LLE based Graph (c) NNK Graph

Fig. 1: Graph constructions in simple scenarios illustrating the sparsity of NNK graphs. Unlike KNN and LLE, NNK achieves a sparse representation where the *relative locations* of nodes in the neighborhood are key. In the seven-node example (top row), the node in the center has only three direct neighbors, which in turn are neighbors of nodes extending the graph further in those three directions. Graphs are constructed with K=5 neighbors and a Gaussian kernel ($\sigma = 1$).

optimization at each node involves solving:

$$\min_{\boldsymbol{\theta}:\ \boldsymbol{\theta}\geq 0} ||\boldsymbol{\phi}_i - \boldsymbol{\Phi}_S\boldsymbol{\theta}||_{l_2}^2. \tag{3}$$

where the goal is to approximate a vector ($\boldsymbol{\phi}_i$) in representation space by a linear combination of atoms (with weights given by $\boldsymbol{\theta}$) from a dictionary obtained from the set S of neighbors ($\boldsymbol{\Phi}_S$).

The selection implied by NNK is equivalent to optimizing $\boldsymbol{\theta}$ (weights) to achieve orthogonality of the residual error, similar to the orthogonalization step in orthogonal matching pursuits [26] (Section 3.1). A key benefit of NNK graphs is their robustness to the choice of sparsity parameters, such as K and ϵ, used in conventional similarity based methods. This is because the number of neighbors that are assigned non-zero weights is not predetermined (by K or ϵ) and instead depends on the relative positions of neighbors of $\boldsymbol{x}_i$ (Fig. 1). Our proposed approach for graph construction, identifies a small number of regression coefficients (as in locality inducing approaches) in terms of an approximation in representation space (as for similarity based approaches).

Related Work: NNK graphs reduce to previously studied LLE based graphs and related Subspace Clustering graphs [27] under a specific choice of *cosine kernel*, where data space remains unaltered. The problem of (3) has been previously studied in the context of kernelized ridge regression [28], kernel subspace clustering [29] or kernelized matrix completion [30] , to name a few. However, we emphasize that an understanding of the objective under non negativity constraints for the specific purpose of *graph construction* or a *geometrical understanding* on irregular manifolds were not explored in the past. Further, we believe that the interpretations we introduce here can shed new light on these related problems. [31] solves a complementary problem where, given a graph, kernels based on the graph are constructed for matrix completion task. This aligns and reinforces our intuition that graphs are structures of similarity defined on a transformed space distinct from observation space.

2. NON NEGATIVE KERNEL REGRESSION GRAPH

For similarities defined as inner products in RKHS, an equivalent representation can be obtained by using a positive definite kernel in the observation space, i.e., $\boldsymbol{\phi}_i^\top\boldsymbol{\phi}_j = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$ generally referred to as *kernel trick* in the literature [25]. Thus, (3) can be rewritten as the minimization of:

$$\begin{aligned}\mathcal{J}_i(\boldsymbol{\theta}) &= \frac{1}{2}||\boldsymbol{\phi}_i - \boldsymbol{\Phi}_S\boldsymbol{\theta}||_{l_2}^2\\ &= \frac{1}{2}(\boldsymbol{\phi}_i - \boldsymbol{\Phi}_S\boldsymbol{\theta})^\top(\boldsymbol{\phi}_i - \boldsymbol{\Phi}_S\boldsymbol{\theta})\\ &= \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{K}_{S,S}\boldsymbol{\theta} - \boldsymbol{K}_{S,i}^\top\boldsymbol{\theta} + \frac{1}{2}\boldsymbol{K}_{i,i},\end{aligned} \tag{4}$$

where $\boldsymbol{K}_{i,j} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Thus, the NNK problem can be written:

$$\boldsymbol{\theta}_S = \underset{\boldsymbol{\theta}:\ \boldsymbol{\theta}\geq 0}{\arg\min}\ \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{K}_{S,S}\boldsymbol{\theta} - \boldsymbol{K}_{S,i}^\top\boldsymbol{\theta}. \tag{5}$$

Equation (5) does not require explicit definition of the transform and requires only knowledge of similarity between observations, represented by the $\boldsymbol{K}$ matrices. Thus, the NNK graph construction framework can be potentially extended to non standard kernels, such as for example similarity metrics learned from data [32].

To complete the graph construction, the i-th row of the adjacency matrix $\boldsymbol{W}$ is given by $\boldsymbol{W}_{i,S} = \boldsymbol{\theta}_S$ and $\boldsymbol{W}_{i,S^c} = \boldsymbol{0}$. Note that each edge weight in S is associated with a local objective value at its optimum (5). We resolve edge weight conflicts for graph symmetrization by choosing the weight associated to smaller objective.

The proposed method consist of two steps. First, we find K nearest neighbors corresponding to each node. Although brute force implementation of this step requires $O(N^2Kd)$, there exist efficient algorithms to find approximate neighbors in $O(N^{1.14})$ [33]. Second, we solve a non-negative kernel regression (5) at each node. This objective is a constrained quadratic function of $\boldsymbol{\theta}$ and can be solved efficiently using a variety of quadratic methods[1] [34] that run in $O(K^3)$ leading to $O(NK^3)$ complexity for this step.

Algorithm 1: Non Negative Kernel regression Algorithm

Input : Kernel $\boldsymbol{K} \in \mathbb{R}^{N\times N}$, No. of Neighbors $\mathcal{K}$
for $i = 1, 2, \ldots N$ **do**
  $S = \{\mathcal{K}$ nearest neighbors of node $i\}$
  $\boldsymbol{\theta}_S = \min_{\boldsymbol{\theta}\geq 0} \frac{1}{2}\boldsymbol{\theta}_S^\top\boldsymbol{K}_{S,S}\boldsymbol{\theta}_S - \boldsymbol{K}_{S,i}^\top\boldsymbol{\theta}_S$
  $\mathcal{J}_i = \frac{1}{2}\boldsymbol{\theta}_S^\top\boldsymbol{K}_{S,S}\boldsymbol{\theta}_S - \boldsymbol{K}_{S,i}^\top\boldsymbol{\theta}_S + \frac{1}{2}\boldsymbol{K}_{i,i}$
  $\boldsymbol{W}_{i,S} = \boldsymbol{\theta}_S, \quad \boldsymbol{W}_{i,S^c} = \boldsymbol{0}$
  $\boldsymbol{E}_{i,S} = \mathcal{J}_i\boldsymbol{1}_{\mathcal{K}}, \quad \boldsymbol{E}_{i,S^c} = \boldsymbol{0}$
end
$\boldsymbol{W} = \mathbb{I}(\boldsymbol{E} \leq \boldsymbol{E}^\top)\odot\boldsymbol{W} + \mathbb{I}(\boldsymbol{E} > \boldsymbol{E}^\top)\odot\boldsymbol{W}^\top$
Output: Graph Adjacency $\boldsymbol{W}$, Local error $\boldsymbol{E}$

3. PROPERTIES OF NNK GRAPHS

Unlike previous graph constructions (e.g., KNN graphs) where an edge connects nodes j and k to node i based solely on the metrics for (i, j) and (i, k), NNK graphs provide a new construction by taking into account, in addition, the relative positions of nodes j and k using the metric on (j, k). In this section, we present an analysis to establish theoretically some of the properties of NNK graphs leading to a geometric interpretation.

[1] In our experiments, we solve the optimization using a modified version of the solver originally implemented as part of nnlslab

3.1. Basis Selection Interpretation

KNN and ϵ neighborhood graphs use the kernel weights to select and assign edge weights with the same kernel weight. Under the distinction of data and transformed space, a kernel value can be interpreted as a projection of one node onto another, i.e., at a node i,

$$k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{\phi}_i^\top \boldsymbol{\phi}_j = \frac{\boldsymbol{\phi}_i^\top \boldsymbol{\phi}_j}{\boldsymbol{\phi}_i^\top \boldsymbol{\phi}_i} \quad \text{since} \quad ||\boldsymbol{\phi}_i||^2 = 1 \tag{6}$$

In the context of sparse dictionary-based representation it is well-known that basis selection via thresholding is suboptimal and can be outperformed by a number of alternative methods, such as matching pursuits (MP) [35] or orthogonal matching pursuits (OMP) [26], among others. In OMP, basis vectors are selected one by one in a greedy fashion as in MP. But, after a new basis vectors has been selected, the coefficients for the representation *in all the selected basis vectors* are recomputed so as to guarantee that the approximation error is orthogonal to the space spanned by the vectors selected.

To see the analogy with our proposed scheme, we first present the steps involved if OMP was to be performed directly on the transformed space, namely,

$$j_1 = \arg\max_j \ \boldsymbol{\phi}_i^\top \boldsymbol{\phi}_j$$

then compute the residue corresponding to approximating $\boldsymbol{\phi}_i$:

$$\boldsymbol{\phi}_{res_1} = \boldsymbol{\phi}_i - \boldsymbol{K}_{i,j_1} \boldsymbol{\phi}_{j_1}$$

so that at step s:

$$j_s = \underset{j \neq j_1, j_2 \cdots j_{s-1}}{\arg\max} \ \boldsymbol{\phi}_j^\top \boldsymbol{\phi}_{res_{s-1}}$$

and denoting:

$$\boldsymbol{\Phi}_S = \left[\boldsymbol{\phi}_{j_1} \boldsymbol{\phi}_{j_2} \dots \boldsymbol{\phi}_{j_s}\right],$$

recomputing the weights associated to each basis (i.e., the edge weights) corresponds to orthogonalizing error with respect to the span of the chosen bases (least squares approximation), leading to:

$$\boldsymbol{w}_s = \min_{\boldsymbol{w}} \ \boldsymbol{\Phi}_S^\top \boldsymbol{\phi}_{res_s} \quad \triangleright \ \boldsymbol{\phi}_{res_s} = \boldsymbol{\phi}_i - \boldsymbol{\Phi}_S \boldsymbol{w}. \tag{7}$$

The above objective can be equivalently achieved by minimizing the energy of the residue obtained as

$$||\boldsymbol{\phi}_{res_s}||^2 = \boldsymbol{\phi}_{res_s}^\top \boldsymbol{\phi}_{res_s} = \boldsymbol{K}_{i,i} - 2\boldsymbol{K}_{S,i}^\top \boldsymbol{w} + \boldsymbol{w}^\top \boldsymbol{K}_{S,S} \boldsymbol{w}.$$

Thus, we see that solving for the weights $\boldsymbol{w}_s$ in (7) by minimizing the residual energy is analogous to the NNK objective (4), but without the constraint of non negativity. The constraints ensure that the graphs obtained are valid with positive edge weights. It is important to note that in most cases the lifting to RKHS space is not explicitly defined, and the space itself is potentially infinite. In such cases, we have access only to the energy of the residue and lack direct knowledge of the residue from OMP. By constraining OMP to positive weights, the span of selected bases at each step belongs to a convex cone while the residue is orthogonal to the cone [36].

The greedy selection step of MP algorithm at each node is presented as Algorithm 2 with stopping condition to ensure positive support to the pursuit problem. The NNK algorithm proposed in Algorithm 1 bypasses this greedy selection with a pre-selected set of *good* atoms, though an exact correspondence with OMP can be made by sequentially selecting neighbors and orthogonalizing to reduce residue's energy. The NNK graph construction algorithm can be interpreted as a computationally efficient method for producing representations with orthogonal approximation errors, which in turn favors sparser representations [26].

Algorithm 2: MP Algorithm

$j_1 = \arg\max_j \ \boldsymbol{K}_{i,j}$
$\boldsymbol{\theta}_1 = \boldsymbol{K}_{i,j_1}, \quad S = \{j_1\}$
for $s = 2, 3, \dots \mathcal{K}$ **do**
  $j_s = \underset{j \notin S}{\arg\max} \ \boldsymbol{K}_{i,j} - \boldsymbol{K}_{S,j}^\top \boldsymbol{\theta}_{s-1}$
  if $\boldsymbol{K}_{i,j_s} - \boldsymbol{K}_{S,j_s}^\top \boldsymbol{\theta}_{s-1} < 0$ **then** break
  $\boldsymbol{\theta}_s^\top = \left[\boldsymbol{\theta}_{s-1}^\top \ \ (\boldsymbol{K}_{i,j_s} - \boldsymbol{K}_{S,j_s}^\top \boldsymbol{\theta}_{s-1})\right], \quad S = S \cup \{j_s\}$
end

3.2. Geometric Interpretation

NNK graph construction solves a constrained optimization at each node, the solution of which corresponds to weights of edges associated to the node. Proposition 1 allows us to analyze edge connections, one pair at a time, as an edge that has weight zero (active constraint) will remain zero at the optimal solution. We introduce mathematical conditions for edges to satisfy for existence in the form of the Kernel Ratio Interval (KRI) theorem, which applied *inductively* unfold the geometric nature of NNK graphs[2].

Proposition 1. *The quadratic optimization problem of (5), solved at each node in NNK framework, satisfies active constraints set*[3] *and can be analyzed in blocks [38].*

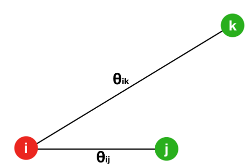


Fig. 2: NNK graph construction at i given nodes j and k.

Theorem 1. ***Kernel Ratio Interval:*** *In a three node scenario (Fig 2), the necessary and sufficient condition for nodes j and k to be connected to i in a NNK graph is given by*

$$\boldsymbol{K}_{j,k} < \frac{\boldsymbol{K}_{i,j}}{\boldsymbol{K}_{i,k}} < \frac{1}{\boldsymbol{K}_{j,k}} \tag{8}$$

Proof. An exact solution to equation (5) under the constraint $\boldsymbol{\theta} \geq \boldsymbol{0}$ is such that

$$\begin{bmatrix} 1 & \boldsymbol{K}_{j,k} \\ \boldsymbol{K}_{j,k} & 1 \end{bmatrix} \begin{bmatrix} \theta_{ij} \\ \theta_{ik} \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}_{i,j} \\ \boldsymbol{K}_{i,k} \end{bmatrix} \tag{9}$$
$$\iff \theta_{ij} + \theta_{ik} \boldsymbol{K}_{j,k} = \boldsymbol{K}_{i,j}$$
$$\theta_{ij} \boldsymbol{K}_{j,k} + \theta_{ik} = \boldsymbol{K}_{i,k}$$

Taking the ratio of the above equations

$$\frac{\theta_{ij} + \theta_{ik} \boldsymbol{K}_{j,k}}{\theta_{ij} \boldsymbol{K}_{j,k} + \theta_{ik}} = \frac{\boldsymbol{K}_{i,j}}{\boldsymbol{K}_{i,k}} \tag{10}$$

[2] A longer version of the paper is posted on arxiv.org for completeness of theoretical statements in this section [37]

[3] In constrained optimization problems, some constraints will be strongly binding, i.e., the solution to optimization at these elements will be zero to satisfy the KKT condition of optimality. These constraints are referred to as **active constraints**, knowledge of which helps reduce the problem size as one can focus on the inactive subset that requires optimization. The constraints that are active at a current feasible solution will remain active at the optimal.

$0 \leq \boldsymbol{K}_{j,k} \leq 1$, where $\boldsymbol{K}_{j,k} = 1$ if and only if the points are same. Without loss of generality, let us assume points j and k are distinct.

$$\boldsymbol{K}_{j,k} < 1 \leq \frac{\boldsymbol{K}_{i,j}}{\boldsymbol{K}_{i,k}} \iff \boldsymbol{K}_{j,k} < \frac{\theta_{ij} + \theta_{ik}\boldsymbol{K}_{j,k}}{\theta_{ij}\boldsymbol{K}_{j,k} + \theta_{ik}}$$

$$\iff \theta_{ij} + \theta_{ik}\boldsymbol{K}_{j,k} > \theta_{ij}\boldsymbol{K}_{j,k}^2 + \theta_{ik}\boldsymbol{K}_{j,k}$$

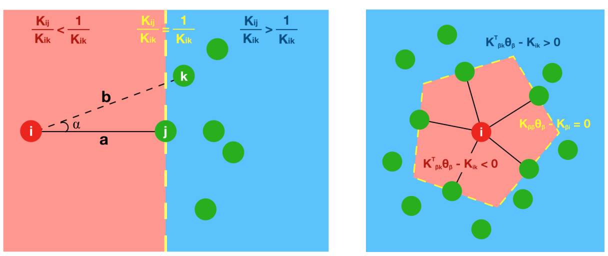
$$\iff \theta_{ij} > \theta_{ij}\boldsymbol{K}_{j,k}^2 \iff \theta_{ij} > 0$$

$$\boxed{\theta_{ij} > 0 \iff \boldsymbol{K}_{j,k} < \frac{\boldsymbol{K}_{i,j}}{\boldsymbol{K}_{i,k}}} \tag{11}$$

Swapping nodes j and k gives the other inequality.

Corollary 1. ***Plane Property:*** *Each edge θ_{ij} in a NNK graph corresponds to a hyper plane with normal in the edge direction (Figure 3a), points beyond which will be not connected for the case of Gaussian kernel (1).*

Corollary 2. ***Polytope Property (Local Geometry of NNK):*** *The local geometry in NNK graph construction, for a given node i, is a convex polytope around the node (Figure 3b).*



(a) (b)

Fig. 3: Local geometry (denoted in red) of NNK graph construction for a Gaussian kernel. a) Plane associated to an edge. b) Convex polytope associated with a node.

4. EXPERIMENTS

We compare our proposed method (NNK, Algorithm 1) and the matching pursuit variant (MP, Algorithm 2) with state of the art method such as Kalofolias [22], AEW [16] and standard KNN and LLE based graphs [19]. Experiments show NNK graphs are robust to choice of K and outperform previous state of the art in graph based semi supervised learning (SSL) on subsets of USPS, MNIST.

Sparsity: Fig. 4 shows the effect of sparsity parameter K on the edge density of various graph methods on synthetic and real data. NNK, MP method produces sparse graphs indicative of their ability to adapt to the local geometry requiring only neighbors that help represent the node in similarity space. LLE graphs, though seemingly sparse in the case of MNIST, suffer from the large dimensionality of the data producing unstable weights ($< 10^{-8}$) and consequently weaker SSL performance. Note that the performance of MP method comes at the cost of longer runtime $\sim$ 2-2.5 times that of NNK.

Semi supervised learning: In this experiment, we evaluate different graphs using SSL method of [6]. For USPS, we folow [6, 23] by non uniformly sampling each class based on its labels ($2.6c^2 \quad c = \{1, 2, \dots 10\}$) producing a dataset of size 1001. For MNIST, we use 100 randomly selected samples for each digit class to create one instance of 1000 sample dataset. A gaussian kernel (1) of bandwidth (σ^2) such that the K neighbors are within 3 standard deviations, is used as the similarity kernel. Fig.5 shows performance (mean over 10 different initializations of available labels) with different graph constructions for various choice of available labels and graph density (K). NNK methods perform best with both combinatorial ($\boldsymbol{L}$) and symmetric normalized Laplacians ($\mathcal{L}$) under most settings.

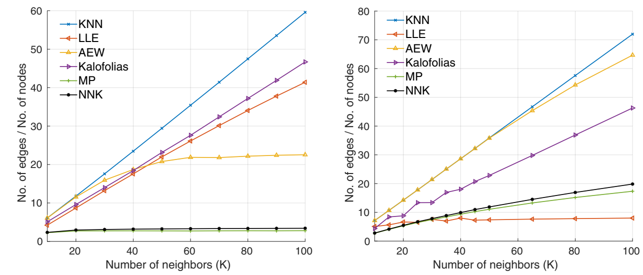


Fig. 4: Ratio of number of edges (above a threshold of 10^{-8}) to the number of nodes for (**Left**) Non uniformly sampled data from Noisy Swiss Roll (N=1000, d=3) and (**Right**) MNIST (N=1000, d=784).

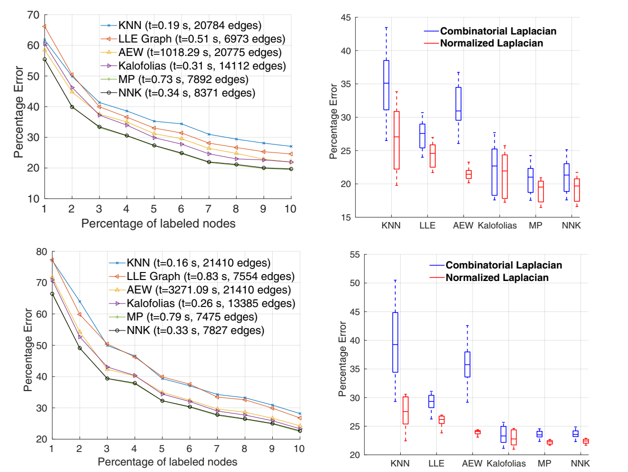


Fig. 5: (**Left column**) Misclassifications at different percentage of labeled data on USPS (**top row**) and MNIST (**bottom row**) dataset with $K = 30$. The time taken and the sparsity of the constructed graphs is indicated as part of the legend. (**Right column**) Boxplots showing the robustness of various graphs for different choices of K $(10, 15, 20, 25, \dots, 50)$ at 10% labeled data on USPS and MNIST.

5. CONCLUSION

We present an efficient and fast framework for graph learning using Non Negative Kernel (NNK) regression with desirable geometrical properties. Observing KNN, ϵ-graphs as naive thresholding methods and LLE based graphs as particular case of NNK allows us to capture earlier graph constructions under a single theoretical model. Future work will explore the connection between NNK's local polytope geometry with the data manifold and scaling to large datasets.

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